Indrajit Kumar

Full-Stack Developer | Product Builder | AI Integration

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Professional Summary

Software Engineer, Full-Stack Developer, and Product Builder passionate about transforming ideas into scalable digital products through AI Integration, modern technologies, problem-solving, and a strong focus on performance, user experience, and innovation.

Education

Jagannath University Jaipur, India

B. Tech in Computer Science Jul 2023 – Expected Jul 2027

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, AI, ML

Technical Skills

- **Languages:** C, C++ (DSA), JavaScript (ES6+), Python, Java, SQL
- **Frontend:** HTML5, CSS3, React.js, Next.js, Vite, Tailwind CSS
- **Backend:** Node.js, Express.js, Django, Flask, FastAPI, GraphQL, Socket.io, WebSocket, REST API, JWT Authentication, Firebase Authentication, Firebase, Supabase, AWS, Google Cloud
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB
- **AI/ML:** AI Integration, Generative AI, Prompt Engineering, LLM Integration, OpenAI API, Google Gemini API, Grok API, Ollama, Llama 3, RAG, Qwen, TensorFlow, PyTorch
- **DevOps & Tools:** Git, GitHub, VS Code, Android Studio, Docker, Firebase, Vercel, Render, Netlify, MySQL Workbench

Projects

PraGo - AI Powered Medical Super App ***Source Code*** **|** ***Live***

React.js & Vite, Tailwind CSS, Lucide React, Axios, Node.js & Express.js, MySQL 8.0, Redis, Groq (Llama 3), Google Gemini Vision, Razorpay, Docker.

- **Built an enterprise-grade multi-tenant** microservices healthcare platform featuring **Dr. PraGo multimodal AI clinical triage** (Groq & Gemini Vision), **Central API Gateway**, and **3 role-based portals** (Patient, Doctor, Admin).
- **Designed** a security-hardened **Strangler Fig architecture** supporting **tamper-proof Razorpay checkout**, **zero-IDOR encrypted medical vaults**, **real-time emergency SOS geo-dispatching**, and **Docker-based deployment**.

Indra-MarketMind - Real-Time Quantitative AI Market Intelligence Terminal ***Source Code*** **|** ***Live***

Next.js 16, TypeScript, Tailwind CSS, Python (FastAPI), PyTorch, Hugging Face (FinBERT, RoBERTa), Prophet, PostgreSQL (pgvector), Redis 7, Docker, Kubernetes.

- **Architected an institutional financial intelligence terminal** featuring a **5-model NLP sentiment ensemble** (FinBERT, RoBERTa, FinGPT**), hybrid Prophet + PyTorch Bi-LSTM** price forecaster with Monte Carlo risk cones, and an **interactive 19 global exchange** session radar.
- **Engineered an asynchronous 10-microservice mesh** backed **by polyglot persistence** (PostgreSQL 16, pgvector, Redis 7), sub-100ms Vercel Edge caching, **CyberShield security** (sliding-window rate limits, strict CSP), and automated paper trading with Kelly Criterion risk controls.

VELOCITY X — 3D Cyber Highway Racing & Pursuit Game ***Source Code*** **|** ***Live***

Three.js (WebGL 2.0), TypeScript, React 18, Vite, Web Audio API, Capacitor 8.5 (Android/iOS), Electron, Service Workers (PWA), Vercel Edge.

- **Architected** a console-grade **3D WebGL** highway pursuit racing engine featuring **PBR automotive clearcoat shaders,** an **8-wave staggered slalom traffic AI** with 32m lookahead anti-clipping distance pacing, aggressive state police interceptors executing physical **PIT maneuvers,** and **a zero-asset procedural Web Audio synthesizer (**RPM-modulated multi-oscillators, turbo blow-off, Doppler sirens).
- **Engineered** a **decoupled dual-loop architecture** maintaining a **locked 60–120 FPS envelope** via **pre-allocated object pooling** (zero GC allocations in render loop), instanced mesh draw-call minimization (<45 calls), a **20Hz-throttled React telemetry bridge**, and sub-**250KB (gzip)** cross-platform distribution spanning **100% offline PWA**, native **Android APK**, and **Vercel Edge**.

Yaadon Ki Duniya — Interactive Audio-Visual Indian Nostalgia Platform ***Source Code*** **|** ***Live***

React 18, TypeScript 5, Vite, Web Audio API (AudioContext, OscillatorNode), HTML5 Canvas, YouTube iFrame API, CSS3 (Glassmorphism), GitHub Actions.

• **Architected an interactive 90s Indian nostalgic streaming platform** featuring a 6-environment ambient soundscape engine, zero-latency procedural Web Audio synthesizer (generating bus horns, scissors, and kettle whistling via math oscillators), and an analog 88.5–107.9 MHz FM radio tuner with synthetic static frequency drift.

• **Engineered a proprietary mathematical 2:1 interleaved playback algorithm** balancing a 700+ song catalog, dynamic dual-layer audio mixer (ambient environmental audio + YouTube music sync), sub-300ms stream error recovery interceptors, and client-side HTML5 Canvas retro postcard generator.

Certifications

- **PCAP: Programming Essentials in Python**
  Cisco Networking Academy & OpenEDG Python Institute ***July 2024*** **|** ***Certificate***
- **Introduction to Cybersecurity**
  Cisco Networking Academy ***June 2025*** **|** ***Certificate***
- **SWOC 2026 – Certificate of Appreciation** ***2026*** **|** ***Certificate***